

JAYRAJ M. PAMNANI

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Work Authorization: F1 OPT (12 Months) | STEM Extension Eligible (36 Months Total)

SUMMARY

Engineer with experience across software engineering, cloud infrastructure, and machine learning systems. Built and deployed scalable applications and production ML pipelines using Kubernetes, Docker, CI/CD, and AWS, with demonstrated impact in infrastructure optimization, deployment automation, and system reliability.

PROFESSIONAL EXPERIENCE

GBCS Group - Software Engineering Intern, Calgary, Canada

11/2025 - 02/2026

- Optimized cloud infrastructure and resource utilization, **reducing hosting and maintenance costs by 28%**.
- Improved CI/CD pipelines and deployment workflows, **accelerating release cycles by 40%** while maintaining 99.9% system availability.
- Collaborated with a team of 5 engineers to build scalable microservices, reducing technical debt and **improving system maintainability by 30%**.

New York University - Teaching Assistant – Machine Learning, New York, NY

09/2025 – 12/2025

- Guided **50+** graduate students through ML fundamentals, including preprocessing pipelining, supervised/unsupervised learning, Deep Learning, and model optimization techniques.
- Conducted weekly office hours to debug Python code, explained algorithms, taught ML topics, and assisted with PyTorch implementations.

Swaroop.ai - AI Intern, Ahmedabad, India

03/2024 - 08/2024

- Enhanced Text-to-Speech model performance by **25%** through **fine-tuning Coqui TTS** across multiple Indian languages, optimizing real-time synthesis pipelines.
- Improved Speech-to-Text accuracy by **18%** by refining Coqui STT models with domain-specific audio datasets and custom preprocessing scripts.
- Integrated **OpenAI Whisper** into the production workflow, increasing transcription accuracy on noisy datasets by **30%** and cutting **inference latency by 20%**.
- Streamlined the **end-to-end MLOps pipeline** (data cleaning, model training, deployment orchestration) to enhance velocity, reducing model turnaround time from **3 days to under 24 hours**.

Freelance Software Developer, Vadodara, India

07/2023 – 09/2023

- Engineered and deployed a Django-based web app that automated data transfer between **QuickBooks and KatanaMRP**, cutting the client's manual bookkeeping workload by **~85%**.
- Built a robust **REST API integration** using Django (backend) and JavaScript (frontend), enabling seamless synchronization of **10,000+ financial and inventory records** with zero data loss.
- Designed and implemented **5+ custom verification layers**, achieving **99.8% data accuracy** before syncing to KatanaMRP.
- Deployed on **AWS EC2** with automated error logging and uptime above **99.9%**, ensuring secure, continuous operation for the client's business processes.

SKILLS

Python, R, Scala, C++, Go • PyTorch, TensorFlow, Scikit-learn, LLMs/RAG • Spark, PostgreSQL, MongoDB, Snowflake • Docker, Kubernetes, Terraform, CI/CD • AWS, GCP, Azure • Django, Flask, Next.js • Kafka, Airflow, MLflow, Tableau, Power BI

EDUCATION

New York University | Master's in Computer Engineering | May 2026

Relevant Coursework: High Performance Machine Learning, Machine Learning Operations, Deep Learning, Database Systems, Big Data

Parul University | Bachelor's in Computer Science & Engineering with Specialization in AI | May 2024

Relevant Coursework: Data Structures and Algorithms, Natural Language Processing, GPU computing, Pattern Recognition

PROJECTS

[ActualBudget Transaction Categorizer](#) | Python, FastAPI, PostgreSQL, MLflow, Docker, Kubernetes, Terraform

- Built an end-to-end MLOps system** that automatically categorizes personal finance transactions inside ActualBudget using a three-layer pipeline: a fastText population classifier, a per-user semantic similarity layer, and a cluster-based custom-category discovery layer that suggests new user-specific categories.
- Designed the full lifecycle around data ingestion, preprocessing, training, evaluation, and serving, with MLflow experiment tracking, Dockerized model services, PostgreSQL-backed persistence, and Kubernetes deployment managed through Terraform, Ansible, Kubespray, Sealed Secrets, and Argo CD for GitOps-style infrastructure control.

[HexDrop \(Secure File Transferring WebApp\)](#) | Next.js, TypeScript, Prisma, PostgreSQL, AWS, Docker, K8s

- Built a secure file-sharing app where users upload files, receive a shareable 6-digit key, and download via that key; implemented end-to-end flow with client-side encryption, Prisma-managed metadata in PostgreSQL, and object storage in AWS S3.
- Designed and deployed a full-stack **DevOps pipeline**: multi-stage Docker builds, **GitHub Actions CI/CD** for build and deploy, **EKS orchestration** with HPA and ALB Ingress, and RDS PostgreSQL with External Secrets for configuration management.

[EyeConnect](#) | WebRTC, Supabase, OpenRouter AI, React, TypeScript

- Developed an end-to-end accessibility platform that connects blind users with sighted volunteers using **WebRTC** for peer-to-peer video streaming and **Supabase Realtime** for instant signaling and volunteer matching; awarded **2nd place at NYU Hacks 2025**.
- Architected a hybrid assistance system integrating **OpenRouter AI** for automated vision descriptions and a **React/TypeScript** frontend with **shadcn/ui** to ensure a high-contrast, accessible UI for low-latency visual support.